

# Seth Baird

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## WORK EXPERIENCE

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- **SMS InfoComm (Wistron Corporation)**

Dallas, TX

*NVIDIA Server Repair Technician*

*May 2024 - August 2024*

- **System Repair and Troubleshooting** Diagnosed, refurbished, repaired, and built NVIDIA HGX 8-GPU units.
- **Hardware Validation and Analysis** Used proprietary diagnostic tools to gain insight into the operational state of motherboard subsystems such as power delivery, signal integrity, and FPGA.
- **Servers and Information Technology** Daily activities depended on the use of Linux, SSH, tmux, Microsoft Office, as well as strong communication and note-taking skills.

- **Jawa.gg**

Remote

*Verified Seller*

*October 2021 - August 2023*

- **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
- **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.
- Top 10 seller in overall performance, recognized by CEO Amanda Stefan for driving Jawa's growth.

## INVOLVEMENT

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- **Purdue Electric Racing - Software Team**

- Developed C code for the STM32-based main module of the racecar alongside core members of the organization
- Reworked the CAN communication backend to use two busses simultaneously for new high-throughput motors
- Utilized testing equipment such as CANable to verify interoperability of new and existing systems

## SKILLS AND PROJECTS

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- **Microcontroller, FPGA, and IoT**

- Leveraged Solidworks, OpenSCAD, and a Raspberry Pi Pico running Rust to create a replica arcade game controller
- Added smart capability to a ceiling fan using an ESP8266 Wi-Fi microcontroller (C), the Blynk API, and relays
- Installed a 3D printer monitoring solution using a Raspberry Pi and camera running Octoprint on Linux (Bash)
- Created interactive circuits in lab based around AVR microcontrollers (C) and Altera FPGAs (AHDL)

- **Software Development**

- Developed an android app to provide data and insights to the driver by integrating with the car's engine computer
- Implemented channel scan into a superheterodyne FM radio using regression and a local maximum search algorithm
- Completed a 2 semester long course on data structures at a vocational school before enrollment in university
- Created over 20 proof-of-concept iOS and Android apps to gain familiarity with Swift and Kotlin
- Studied OOP in Java at home and vocational school beginning in 2018, completing 3 courses in the subject
- Leveraged Node.js, Discord.js, and Dotenv to develop an audio capable bot written in Typescript

- **Linux Server Administration**

- Designed custom bash scripts utilizing cron to automate ZFS dataset replication and other maintenance tasks
- Administrated a network of KVM virtual machines and Linux containers (LXC) under the Proxmox hypervisor
- Configured a DNS server with domain filtering to block advertisements and malicious sites in a home network
- Assembled a pair of multipurpose storage servers configured to make bidirectional encrypted backups over VPN

## EDUCATION

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- **Purdue University**

West Lafayette, Indiana

*In progress - Anticipated Graduation May 2027*

*GPA 3.01*

**Major in Computer Engineering Technology Minor in Computer and Information Technology**

- **Memorial High School**

Frisco, Texas

*Graduated May 2023*

*Unweighted GPA 3.81*

- **Vocational education:** CS3: Data Structures, Mobile Application Development